

Is rule-following instrumental to the outcome?

Rules outcome-INSTRUMENTAL

Following the rules is the path
to task success.

chains: RLVP 336 vs GRPO 2240
episodes to 50% (6.7x faster)

gated: RLVP+mix 1 vs
outcome-only 0 (wall broken)

RLVP wins

Rules ORTHOGONAL to outcome

Compliance does not advance
the task objective.

τ 2: outcome reward 0.60
RLVP reward 0.45

process reward pulls policy into a
compliant-but-useless attractor

RLVP collapses